

Eren Dogan

<https://edogan.us>
erdogan@ucsc.edu

Student Researcher at Google — PhD Candidate in CSE at UCSC — VLSI/EDA & Graph ML
5+ years of experience in hardware research & programming

EXPERIENCE

Student Researcher

Google

April 2026 – Present

- Implemented global gate sizing for OpenROAD, an open-source RTL-to-GDS Flow

Graduate Student Researcher

University of California, Santa Cruz

Advisor: Prof. Matthew Guthaus

September 2022 – Present

- **GNN- C_{eff} : Effective Capacitance Modeling Using Graph Neural Networks**
 - Implemented a graph attention network-based effective capacitance modeling
 - Achieved 15x speed-up on real-world benchmarks with 0.01 fF mean absolute error
 - Accepted to ISQED 2026
- **GAT-Steiner: Rectilinear Steiner Minimal Tree Prediction Using GNNs**
 - Developed a graph attention network model to predict RSMT solutions of any degree
 - Achieved 99.85% accuracy on the ISPD19 benchmark
 - Accepted to ICCAD 2024
- **Router for OpenRAM**
 - Implemented a new graph-based, BVH-accelerated supply and signal router for OpenRAM, an open-source SRAM compiler
 - Obtained $1.37\times$ speed-up over the old router

Research Intern

Ozyegin University

Advisor: Prof. H. Fatih Ugurdag

October 2020 – September 2021

- **OpenCache**
 - Developed an open-source custom cache generator using OpenRAM's SRAM arrays
- **Deep Compression for PyTorch Models**
 - Implemented pruning and quantization applied on PyTorch models
 - Achieved decreased memory usage and improved speed with compressed sparse column (CSC) format

EDUCATION

University of California, Santa Cruz

PhD in CSE; GPA: 3.86/4.00

Santa Cruz, CA

September 2022 – June 2027 (expected)

Ozyegin University

BS in CS; GPA: 4.00/4.00

Istanbul, Turkey

September 2017 – June 2022

Honors: Valedictorian, Full-tuition scholarship

PROGRAMMING SKILLS

Languages: Python, C/C++, (System)Verilog, Chisel, Tcl, Java, JavaScript, SQL, Bash

Tools & Frameworks: Linux, Git, PyTorch, PyTorch Geometric, Tensorflow, Spektral, Amaranth, Makefile

MISCELLANEOUS

Award: CITRIS Costas John Spanos Semiconductor Manufacturing Award 2026

Reviewer: DAC 2025, MLCAD 2024, VLSI-SoC 2024

Volunteer: MLCAD 2025

Member: ACM, IEEE